AnalogOne

Technical Datasheet
20W CLASS AB AUDIO AMPLIFIER

Discrete-Component Audio Amplifier

8 Ω Loudspeaker Application

General Description

The AnalogOne Class AB Audio Amplifier is a 20W mono audio amplifier designed to increase audio signal amplitude of a standard music player device with a 3.5mm Jack connector audio to power an 8 Ω nominal impedance Loudspeaker. The main quality feature is the capacity to maintain the audio signal with a THD of 0.88% consider a good/moderate distortion useful for this application. The Amplifier runs off nominal symmetrical ±24 V power supplies that ensure the power delivery to the speaker. The amplifier consists of three functional stages: a BJT preamplifier, a MOSFET voltage amplification stage, and a complementary MOSFET Class AB power output stage.

Key Features

- 20W with 0.88% THD Output into a 8 Ω Load at $1kHz$.
- 8 Ω nominal loudspeaker load.
- Adjustable Volume Control.
- Adjustable MOSFET bias.

Applications

- Multi-Media Speaker System.
- Music Amplifier.
- Analog Audio Module.
- Analog Sound Drive.

Functional Block Diagram



Figure 1: Functional Block Diagram of the Device

Device Circuit Architecture

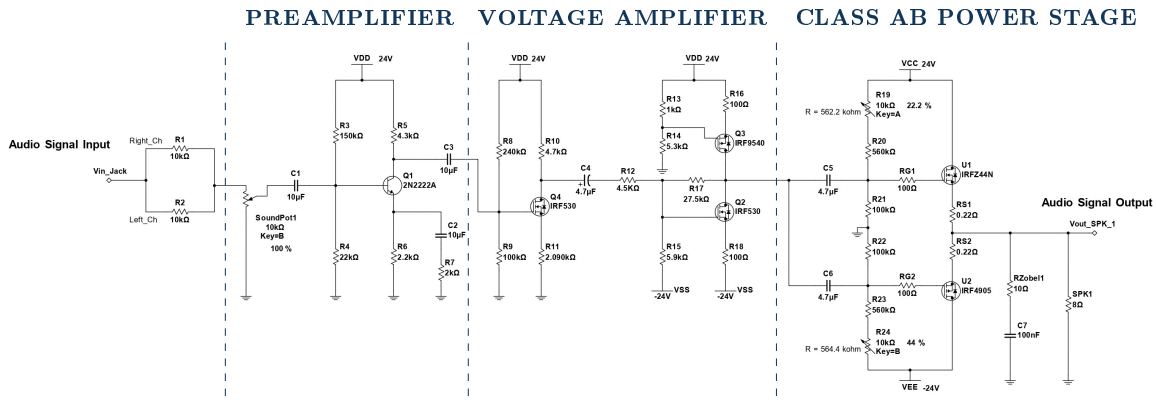


Figure 2: Device Circuit Architecture Modular Division

Component Summary

Component Type	Component	Function
Preamplifier Stage		
NPN Transistor	2N2222	Common-emitter voltage amplification
Voltage Amplification Stage		
N-Channel MOSFET	IRF530	Voltage amplification
P-Channel MOSFET	IRF9540	Active load for the voltage amplification stage
Class AB Power Stage		
N-Channel MOSFET	IRFZ44N	Positive output current delivery
P-Channel MOSFET	IRF4905	Negative output current delivery
Source Resistor	0.22 Ω, 5 W	Current stabilization and source degeneration
Gate Resistor	100 Ω	Limits gate current and improves MOSFET stability
Bias Trimmer	10 kΩ	Adjustment of the Class AB quiescent bias
Coupling Capacitor	4.7 μF	Blocks DC while allowing the audio signal to pass
Output Network		
Zobel Resistor	10 Ω, 2 W	Output stabilization at high frequencies
Zobel Capacitor	100 nF	Forms the Zobel stabilization network with the 10 Ω resistor
Load		
Loudspeaker	8 Ω, 30 W	Nominal audio output load

Table 1: Main amplifier components organized by functional stage of the device

Electrical Specifications

General Electrical Specifications of the Device		
Parameter	**Value**	**Unit**
Supply voltage	±24	V
Current Requiered	–	A
Power Consumption	–	W
RMS input voltage	1–2	V_{RMS}
Max RMS output voltage	12.65	V_{RMS}
Max RMS output current	1.58	A_{RMS}
Frecuency Range	40–20k	Hz
Total harmonic distortion (THD) at 1 kHz	0.88	%
Reference output power	20	W
Power efficiency	–	%
Nominal load resistance	8	Ω

Voltage Amplification Stage

Two IRF530 transistors amplify the preamplifier signal. The IRF9540 acts as the load for the second IRF530. An output capacitor blocks the DC bias while passing the audio signal to the Class AB stage. The design target is 20 V peak for a 2 V RMS input.

Stage characteristic	**Value**	**Unit**
Supply rails	±24	V
Test load	43	kΩ
Small-signal gain near 20 Hz	6.930	V/V
Small-signal gain near 1 kHz	7.118	V/V
Peak-to-peak output at 1 kHz	39.975	V
Output fundamental at 1 kHz	20.112	V peak
Stage THD at 1 kHz (through harmonic 9)	1.183	%

Power and Thermal Characteristics

Component	**Device**	**Power Dissipation**	$R_{\theta JC}$
Preamplifier Stage			
BJT Transistor	2N2222	17.60 mW	83.3°C/W
Voltage Amplification Stage			
N-Channel MOSFET (Q4)	IRF530	20.8 mW	1.60 °C/W
N-Channel MOSFET (Q1)	IRF530	72 mW	1.60 °C/W
P-Channel MOSFET (Q2)	IRF9540	90 mW	1.0 °C/W
Class AB Power Stage			
N-Channel MOSFET	IRFZ44N	6.58 W	1.5 °C/W
P-Channel MOSFET	IRF4905	6.63 W	0.75 °C/W
Output Stage Total	–	**13.21 W**	–

Table 2: Power dissipation and thermal characteristics of the amplifier stages.
Voltage-stage dissipation values are estimates at zero input. Junction-to-case thermal resistances are from the onsemi IRF530/D and Vishay IRF9540 (document 91078) datasheets.

13. Performance Verification

The documented electrical results were obtained through circuit simulation.

The Class AB power-stage reference simulation used a 10 kHz input signal and an 8 Ω load.

The recorded average output power was 22.8 W.

In the standalone voltage-stage simulation, a 2 V RMS sine wave at 1 kHz produced peaks of +19.919 V and −20.056 V across the 43 kΩ test load. The 39.975 V peak-to-peak output corresponds to a gain of 7.07 V/V. Fourier analysis of the same output gave 1.183% total harmonic distortion through the ninth harmonic.

These results correspond to individual stage evaluations.

Further testing is necessary to determine the complete amplifier frequency response, harmonic distortion, noise, and continuous output-power rating.

14. Document Information

Developer	Analog One
Product	Class AB Audio Amplifier
Institution	Tecnologico de Costa Rica
Department	School of Electronics Engineering
Document	Technical Datasheet
Status	Design and Simulation

ANALOG ONE – AUDIO AMPLIFICATION SYSTEM